

Study Tracker App

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① Technology :- Java

② Project overview :-

The Study Tracker App is a Java application designed to help students systematically log, track, summarize and export their study activities.

- It allow users to maintain daily study records, view summarise grouped by date or subject and export all logs into a csv file for offline reference.

- This project demonstrates practical usage of Java collections, File I/O, and object Oriented Design in a real-world, utility driven application.

③ Key Features :-

i) - Insert study log

① Record study session with date (auto-generated), subject, duration and description.

ii) - Display Logs

① view all study logs currently stored in memory.

iii) - summary by date

① calculate and display total study hours grouped by date.

iv) Summary by Subject

- calculate and display total study hours grouped by Date.

v) Export to CSV

- Export all study logs into a CSV file for offline tracking.

vi) user friendly console menu

- Menu driven interface with switch case navigation for ease of use.

○ Technologies used

- Language : Java

- Packages & APIs

- java.util.* → Data Structures (ArrayList, TreeMap), Scanner (input).

- java.time.LocalDate → Auto captures the current date for study log.

- java.io → File handling and CSV export.

Project Flow

Launch the application → Main Menu displayed



choice 1 : Insert new study log



user provides subject, duration, description. Date (auto-generated).

choice 2 : Display all study logs stored in memory.

choice 3 : Display Summary grouped by date (total hrs per day).

choice 4 : Display Summary grouped by subject (total hrs per subject).

choice 5 : Export all study logs to (xyz) file.

choice 6 : Exit application.

① According to user choice, the function gets called and operations are performed.

① Classes & Responsibilities

- StudyLog :-

① Represents a single study session.

① Attributes : Local Date date,
String subject,
double duration,
String description.

① Methods : Constructor , getters , toString();

- StudyTracker

① Manages all logs in memory.

① Attributes : ArrayList <> <StudyLog> database.

① Methods : InsertLog(), DisplayLog(),
SummaryByDate(), Export(svl),
SummaryBySubject().

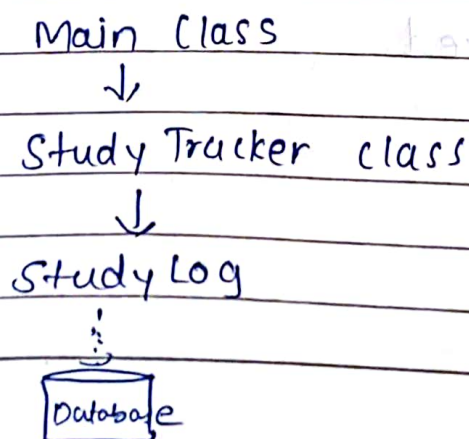
- Study Tracker App (main class)

① Contains main() method.

① Handles menu-driven interface and user input.

① calls appropriate methods from StudyTracker.

① Flow :-



① How to present in an Interview

- I developed a console/GUI - based Java application called the Study-Tracker App
- It allows students to log daily study sessions with subject, duration and description.
- The application can display all logs, generate summaries by Date & Subject, and export records into a csv file for offline use.
- It is built entirely in Core Java using collections (ArrayList, TreeMap), File I/O, and LocalDate API.
- This project improved my skills in Java OOP, data structures, file handling & menu-driven application development.

Java Code

```
1  //////////////////////////////////////
2  //
3  // Project Name : STUDY_TRACKER_APP
4  //
5  //////////////////////////////////////
6
7  import java.io.*;
8  import java.util.*;
9  import java.time.LocalDate;
10 import java.time.LocalDateTime;
11
12 class StudyLog
13 {
14     private LocalDate Date; //characterstics to store Date
15     private String Subject; //subject
16     private double Duration; //Duration
17     private String Description; //Description
18
19     public StudyLog(LocalDate a, String b, double c, String d) //constructor
20     {
21         this.Date = a; //allocate all resources
22         this.Subject = b;
23         this.Duration = c;
24         this.Description = d;
25     }
26
27     @Override
28     public String toString() //we overrides toString() method from Object class(in
built)
29     {
30         return Date + " | " + Subject + " | " + Duration + " | " + Description;
31     }
32
33     public LocalDate getDate() //method to get Date
34     {
35         return this.Date;
36     }
37
38     public String getSubject() //method to get Subject
39     {
40         return this.Subject;
41     }
42
43     public double getDuration() //method to get Duration(study hours)
44     {
45         return this.Duration;
46     }
47
48     public String getDescription() //method to get Description (info)
49     {
```

```

50         return this.Description;
51     }
52 }
53
54 class StudyTracker
55 {
56     public ArrayList <StudyLog> Database; //Collection framework stores StudyLog class
as Its type
57
58     public StudyTracker()
59     {
60         Database = new ArrayList<StudyLog>(); //initialized ArrayList ,Database holds
StudyLog Object
61     }
62
63     //////////////////////////////////////
64     //
65     // Function Name :      InsertLog()
66     // Description :      It is used to Insert new Logs
67     // Author :      Darshan Ananta Thorbole
68     // Date :      21/08/2026
69     //
70     //////////////////////////////////////
71     public void InsertLog()
72     {
73         Scanner sobj = new Scanner(System.in);
74
75         System.out.println("-----");
76         System.out.println("---- Enter the details of your study ----");
77         System.out.println("-----");
78
79         LocalDate lobj = LocalDate.now(); // inbuilt methods gives local time of
system automatically(API)
80
81         System.out.println("We are entering the date as : "+lobj);
82
83         System.out.println("Enter the name of subject like C/C++/Java etc");
84         String sub = sobj.nextLine(); //Stores name of object
85
86         System.out.println("Enetr the time period of your study : ");
87         double dur = sobj.nextDouble(); //Stores duration/time
88
89         sobj.nextLine(); //it is IMP to clean Bufferedreader
90
91         System.out.println("Please provide the description of your study : ");
92         String desc = sobj.nextLine(); //It stores description
93
94         StudyLog studyobj = new StudyLog(lobj, sub, dur, desc); //calls StudyLog
Constructor
95
96         Database.add(studyobj); //.add() adds data to StudyLog
97
98         System.out.println("Study log gets inserted succesfully");
99

```

```
100         System.out.println("-----");
101     }
102
103     //////////////////////////////////////
104     //
105     // Function Name :      DisplayLog()
106     // Description :      It is used to Display all Log
107     //                   details on screen.
108     // Author :           Darshan Ananta Thorbole
109     // Date :             21/08/2026
110     //
111     //////////////////////////////////////
112     public void DisplayLog()
113     {
114         System.out.println("-----");
115
116         if(Database.isEmpty()) //filter
117         {
118             System.out.println("Nothing to display - database is empty");
119             System.out.println("-----");
120
121             return;
122         }
123
124         System.out.println("-Log Report of Marvellous Study Tracker- ");
125         System.out.println("-----");
126
127         for(StudyLog s : Database) //using for-each loop
128         {
129             System.out.println(s);
130         }
131
132         System.out.println("-----");
133
134     }
135     //////////////////////////////////////
136     //
137     // Function Name :      DeleteLog()
138     // Description :      It is used to delete an
139     //                   incorrectly inserted study log.
140     // Author :           Darshan Ananta Thorbole
141     // Date :             22/08/2026
142     //
143     //////////////////////////////////////
144     public void DeleteLog()
145     {
146         Scanner sobj = new Scanner(System.in);
147
148         System.out.println("-----");
149
150         if(Database.isEmpty()) //filter
151         {
152             System.out.println("Nothing to delete - database is empty");
153             System.out.println("-----");
```



```

154         return;
155     }
156
157     System.out.println("----- Study Logs -----");
158
159     for(int i = 0; i < Database.size(); i++)
160     {
161         System.out.println((i + 1) + " : " + Database.get(i)); //provides index for
each file
162     }
163
164     System.out.println("-----");
165     System.out.println("Enter the log number that you want to delete : ");
166
167     int iChoice = sobj.nextInt();
168
169     if(iChoice < 1 || iChoice > Database.size()) //filters
170     {
171         System.out.println("Invalid log number");
172         System.out.println("-----");
173         return;
174     }
175
176     Database.remove(iChoice - 1); //use to delete log
177
178     System.out.println("Study log deleted successfully");
179     System.out.println("-----");
180 }
181
182 ///////////////////////////////////////////////////
183 //
184 // Function Name :      ExportToCSV()
185 // Description :      It is used to Export Log data
186 //                    to CSV(Excel File) on HDD.
187 // Author :          Darshan Ananta Thorbole
188 // Date :            21/08/2026
189 //
190 ///////////////////////////////////////////////////
191 public void ExportToCSV()
192 {
193     Scanner sobj = new Scanner(System.in);
194
195     System.out.println("Enter the name that you want to create for CSV file");
196     String FileName = sobj.nextLine();
197
198     System.out.println("-----");
199
200     if(Database.isEmpty())
201     {
202         System.out.println("Nothing to export - database is empty");
203         System.out.println("-----");
204
205         return;
206     }

```

```

207
208     try(FileWriter fwobj = new FileWriter(FileName))
209     {
210         fwobj.write("Date,Subject,Duration of study,Description of study\n");
211
212         for(StudyLog s : Database)
213         {
214             fwobj.write(s.getDate()+","+
215                 s.getSubject()+","+
216                 s.getDuration()+","+
217                 s.getDescription()+"\n");
218         }
219
220         System.out.println("Data gets exported to CSV succesfully");
221
222         System.out.println("-----");
223     }
224     catch(IOException iobj)
225     {
226         System.out.println(iobj);
227     }
228     catch(Exception eobj)
229     {
230         System.out.println(eobj);
231     }
232 }
233
234 ///////////////////////////////////////////////////////////////////
235 //
236 // Function Name :      ImportFromCSV()
237 // Description :      It is used to Import study
238 //                    logs from CSV file into
239 //                    primary memory.
240 // Author :          Darshan Ananta Thorbole
241 // Date :            22/08/2026
242 //
243 ///////////////////////////////////////////////////////////////////
244
245 public void ImportFromCSV()
246 {
247     Scanner sobj = new Scanner(System.in);
248
249     System.out.println("-----");
250     System.out.println("Enter the name of CSV file to import : ");
251
252     String FileName = sobj.nextLine(); //stores file name
253
254     System.out.println("-----");
255
256     int iCount = 0;
257
258     try(BufferedReader brobj = new BufferedReader(new FileReader(FileName)))
259     {
260         // Skip CSV header(reads header line but dont store it)

```

```

261         brobj.readLine();
262
263         String Line = null;
264
265         while((Line = brobj.readLine()) != null) //Stores one line data into Line
Variable
266         {
267             if(Line.trim().isEmpty()) //avoids empty line
268             {
269                 continue; //skips this loop iteration only
270             }
271
272             String Data[] = Line.split(",", -1); //Tokens of word from Line (-1
preservs last col)
273
274             if(Data.length != 4)
275             {
276                 System.out.println("Invalid record skipped : " + Line);
277                 continue;
278             }
279
280             LocalDate Date = LocalDate.parse(Data[0].trim()); //conversion of
datatypes
281
282             String Subject = Data[1].trim(); //.trim() removes before and after spaces
283
284             double Duration = Double.parseDouble(Data[2].trim());
285
286             String Description = Data[3].trim();
287
288             StudyLog studyobj = new StudyLog(Date, Subject, Duration, Description);
//creates new object
289
290             Database.add(studyobj); //inserts data into Database
291
292             iCount++; //counts no.of new studylog added
293         }
294
295         System.out.println(iCount + " study logs imported successfully");
296
297         System.out.println("-----");
298     }
299     catch(FileNotFoundException fobj) //handles exception if occurred to file
operations
300     {
301         System.out.println("CSV file not found");
302     }
303     catch(IOException iobj)
304     {
305         System.out.println("Error while reading CSV file");
306     }
307     catch(Exception eobj)
308     {
309         System.out.println("Invalid data found in CSV file");

```



```

310     }
311 }
312
313 ///////////////////////////////////////////////////
314 //
315 // Function Name :      SummaryByDate()
316 // Description :      It is function like mimic of
317 //                   group by clause of SQL.(Sorts)
318 //                   It groups data by Date.
319 // Author :          Darshan Ananta Thorbole
320 // Date :            22/08/2026
321 //
322 ///////////////////////////////////////////////////
323 public void SummaryByDate()
324 {
325     System.out.println("-----");
326     System.out.println(" Summary by Date from study tracker ");
327     System.out.println("-----");
328
329     TreeMap <LocalDate, Double> tobj = new TreeMap<LocalDate,Double>();
330
331     LocalDate lobj = null;
332
333     double d = 0.0;
334
335     double old = 0.0;
336
337     for(StudyLog s : Database)
338     {
339         lobj = s.getDate();
340         d = s.getDuration();
341
342         if(tobj.containsKey(lobj))
343         {
344             old = tobj.get(lobj);
345             tobj.put(lobj, d+old);
346         }
347         else
348         {
349             tobj.put(lobj, d);
350         }
351     }
352
353     // Display the details as per date
354     for(LocalDate l : tobj.keySet())
355     {
356         System.out.println("Date : "+l+" Total Study Duration : "+tobj.get(l));
357     }
358
359     System.out.println("-----");
360 }
361
362 ///////////////////////////////////////////////////
363 //

```

```

364 // Function Name :      SummaryBySubject()
365 // Description :      It is function like mimic of
366 //                      group by clause of SQL.(Sorts)
367 //                      It groups data by Subject.
368 // Author :           Darshan Ananta Thorbole
369 // Date :             22/08/2026
370 //
371 //////////////////////////////////////
372 public void SummaryBySubject()
373 {
374     System.out.println("-----");
375     System.out.println(" Summary by Subject from study tracker ");
376     System.out.println("-----");
377
378     TreeMap <String, Double> tobj = new TreeMap<String,Double>();
379
380     String sobj = null;
381
382     double d = 0.0;
383
384     double old = 0.0;
385
386     for(StudyLog s : Database)
387     {
388         sobj = s.getSubject();
389         d = s.getDuration();
390
391         if(tobj.containsKey(sobj))
392         {
393             old = tobj.get(sobj);
394             tobj.put(sobj, d+old);
395         }
396         else
397         {
398             tobj.put(sobj, d);
399         }
400     }
401
402     for(String str : tobj.keySet())
403     {
404         System.out.println("Subject : "+str+" Total Study Duration :
405 "+tobj.get(str));
406     }
407     System.out.println("-----");
408 }
409 }
410
411
412
413 //////////////////////////////////////
414 //
415 // Entry Point function of the project
416 //

```

```
417 //////////////////////////////////////////////////
418 class Study_Tracker_App
419 {
420     public static void main(String A[])
421     {
422         int iChoice = 0;
423
424         StudyTracker stobj = new StudyTracker();
425         Scanner sobj = new Scanner(System.in);
426
427         System.out.println("-----");
428         System.out.println("-----Welcome to Study Tracker App -----");
429         System.out.println("-----");
430
431         do
432         {
433             System.out.println("-----");
434             System.out.println("Please select appropriate option : ");
435             System.out.println("-----");
436
437             System.out.println("1 : Insert new study log");
438             System.out.println("2 : View all study logs");
439             System.out.println("3 : Delete study logs");
440             System.out.println("4 : Export study log to CSV");
441             System.out.println("5 : import study log from CSV");
442             System.out.println("6 : Summary of study log by date");
443             System.out.println("7 : Summary of study log by subject");
444             System.out.println("8 : Exit the application");
445
446             System.out.println("-----");
447
448             iChoice = sobj.nextInt();
449
450             switch(iChoice)
451             {
452                 // Insert new log
453                 case 1:
454                     stobj.InsertLog();
455                     break;
456
457                 // View all study logs
458                 case 2:
459                     stobj.DisplayLog();
460                     break;
461
462                 // Delete specific study logs
463                 case 3:
464                     stobj.DeleteLog();
465                     break;
466
467                 // Export to CSV
468                 case 4:
469                     stobj.ExportToCSV();
470                     break;
```

```
471
472         // Import studylog from CSV
473         case 5:
474             stobj.ImportFromCSV();
475             break;
476
477         // Summary by date
478         case 6:
479             stobj.SummaryByDate();
480             break;
481
482         // Summary by subject
483         case 7:
484             stobj.SummaryBySubject();
485             break;
486
487         // Terminate the project
488         case 8:
489             break;
490
491         default:
492             System.out.println("Please enter valid option");
493             break;
494     }
495
496     }while(iChoice != 8);
497
498     System.out.println("-----");
499     System.out.println("--- Thank for using Study Tracker App ---");
500     System.out.println("-----");
501
502     }
503 }
```